

WEEK 7

REACT

HANDS ON 1:

Objectives

- List the features of ES6
- Explain JavaScript let
- Identify the differences between var and let
- Explain JavaScript const
- Explain ES6 class fundamentals
- Explain ES6 class inheritance
- Define ES6 arrow functions
- Identify set(), map()

Feature	Description	Usage
let and const	Used to declare variables with block scope	Used for player arrays
map()	Iterates and transforms array data	Display players and scores
arrow functions	Shorter function syntax	Filtering players with low scores
destructuring	Extract array values	Odd and even index players
spread operator (...)	Merges arrays	Merge T20 and Ranji player arrays
if-else	Conditional rendering	Toggle between components

src/ListofPlayers.js

```
import React from 'react';
```

```
const ListofPlayers = () => {  
  const players = [  
    { name: 'Virat Kohli', score: 80 },  
    { name: 'Rohit Sharma', score: 65 },
```

```
{ name: 'MS Dhoni', score: 90 },
{ name: 'KL Rahul', score: 45 },
{ name: 'Shikhar Dhawan', score: 70 },
{ name: 'Ravindra Jadeja', score: 75 },
{ name: 'R Ashwin', score: 60 },
{ name: 'Bumrah', score: 95 },
{ name: 'Shami', score: 50 },
{ name: 'Hardik Pandya', score: 55 },
{ name: 'Surya Kumar', score: 77 },
];
```

```
const filteredPlayers = players.filter(player => player.score < 70); // ES6 arrow function
```

```
return (
  <div>
    <h2>All Players</h2>
    <ul>
      {players.map((player, index) => (
        <li key={index}>{player.name} - {player.score}</li>
      ))}
    </ul>

    <h2>Filtered Players (Score below 70)</h2>
    <ul>
      {filteredPlayers.map((player, index) => (
        <li key={index}>{player.name} - {player.score}</li>
      ))}
    </ul>
  </div>
);
};
```

```
export default ListofPlayers;
```

src/IndianPlayers.js

```
import React from 'react';

const IndianPlayers = () => {

  const T20players = ['Rohit', 'Kohli', 'Bumrah'];
  const RanjiTrophy = ['Pujara', 'Saha', 'Jadeja'];

  const allPlayers = [...T20players, ...RanjiTrophy]; // ES6 Spread/Merge

  // Destructuring and filtering odd/even index players
  const oddPlayers = allPlayers.filter((_, index) => index % 2 !== 0);
  const evenPlayers = allPlayers.filter((_, index) => index % 2 === 0);

  return (
    <div>
      <h2>All Merged Players</h2>
      <ul>
        {allPlayers.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>

      <h3>Even Team Players</h3>
      <ul>
        {evenPlayers.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>

      <h3>Odd Team Players</h3>
      <ul>
        {oddPlayers.map((player, index) => (
          <li key={index}>{player}</li>
        ))}
      </ul>
    </div>
  );
}
```

```
        </ul>
      </div>
    );
  };

export default IndianPlayers;
```

App.js:

```
import React from 'react';
import ListofPlayers from './ListofPlayers';
import IndianPlayers from './IndianPlayers';

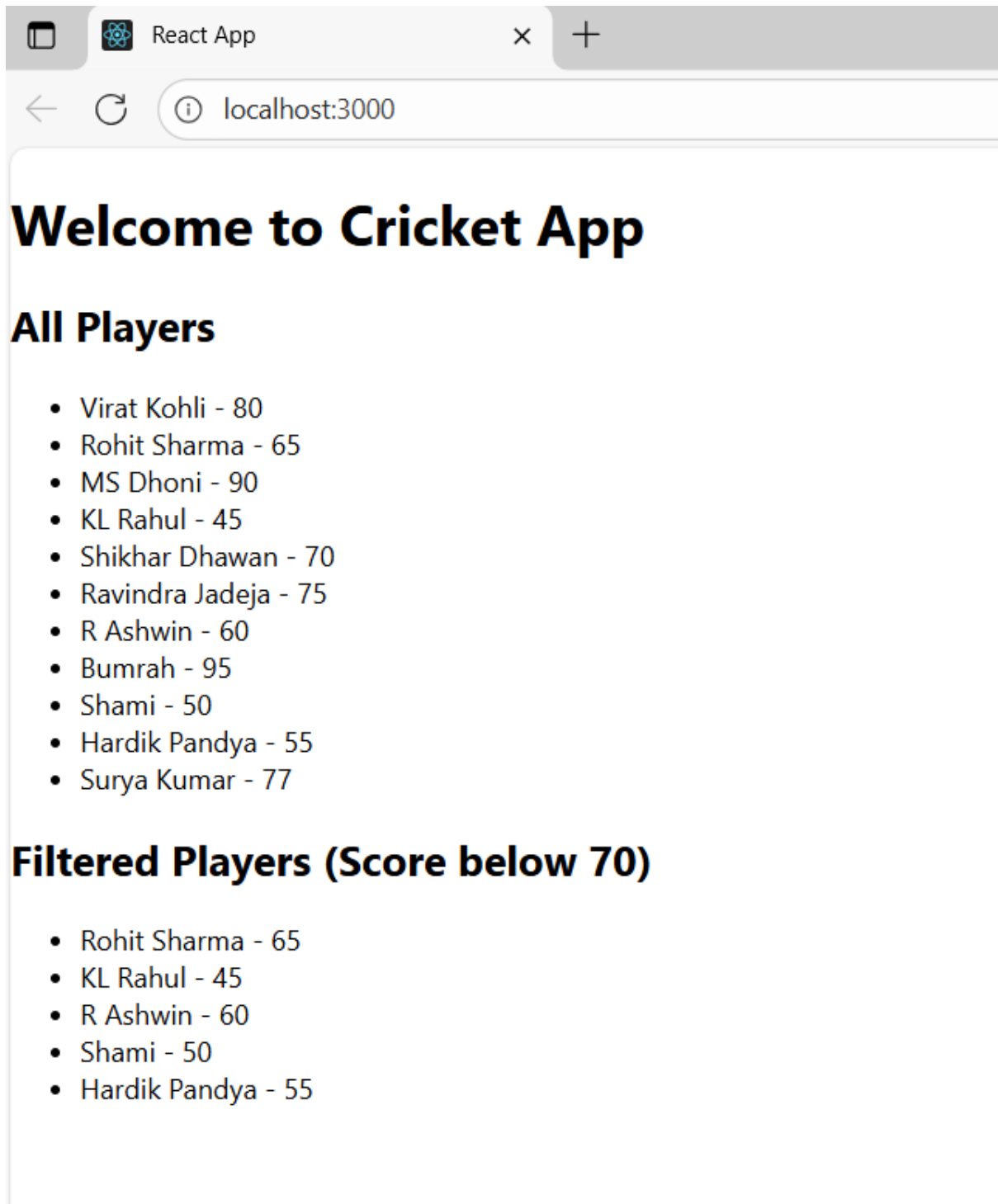
function App() {
  const flag = true; // Change to false to switch components

  return (
    <div className="App">
      <h1>Welcome to Cricket App</h1>
      {flag ? <ListofPlayers /> : <IndianPlayers />}
    </div>
  );
}

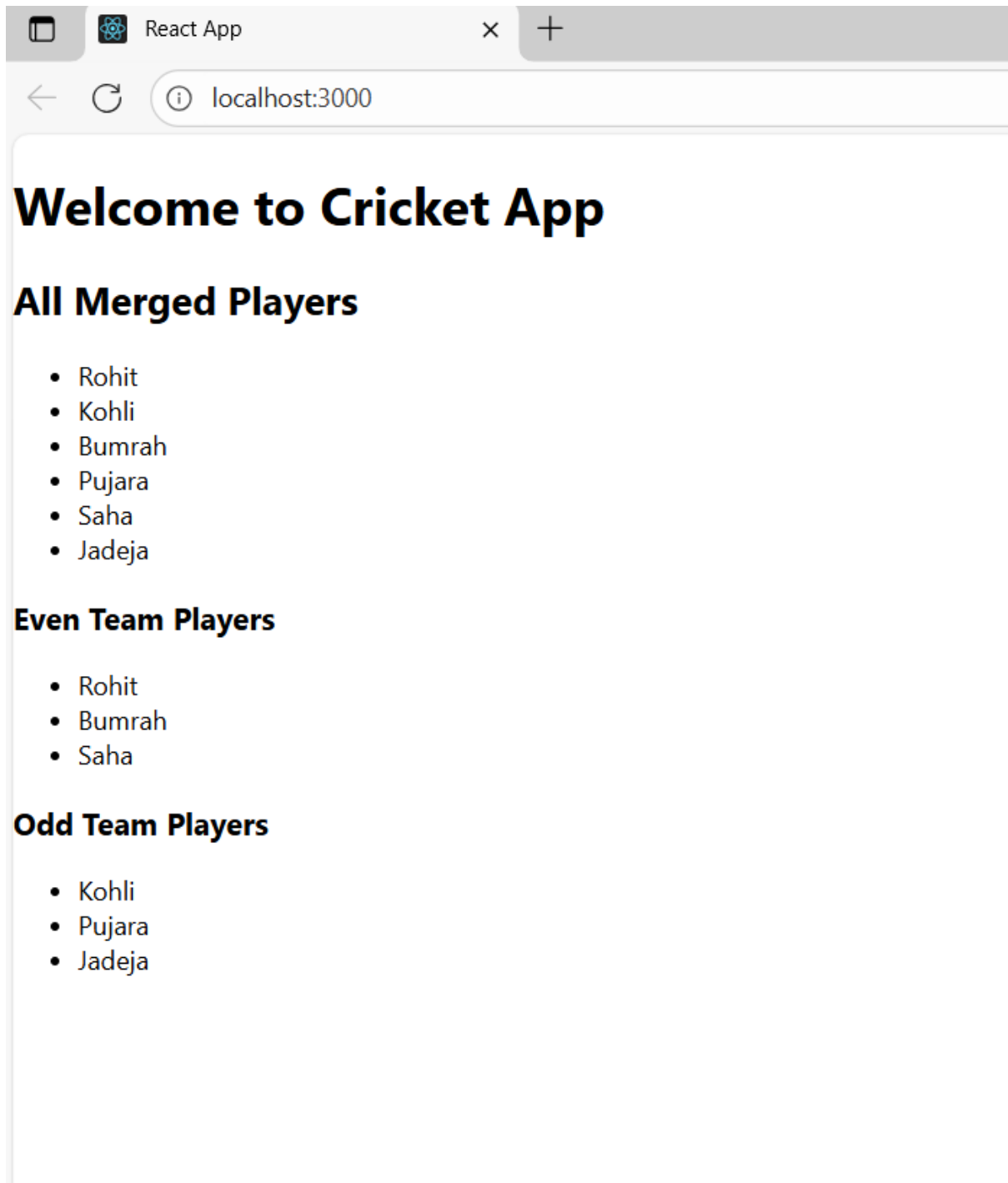
export default App;
```

OUTPUT:

Flag=true



Flag=false:



HANDS ON 2:

Objectives

- Define JSX
- Explain about ECMA Script
- Explain React.createElement()
- Explain how to create React nodes with JSX
- Define how to render JSX to DOM
- Explain how to use JavaScript expressions in JSX
- Explain how to use inline CSS in JSX

App.js:

```
import React from 'react';
```

```
function App() {
```

```
  // Heading JSX element
```

```
  const heading = <h1> 🏢 Office Space Rental App</h1>;
```

```
  // Image Attribute
```

```
  const imageURL = "https://officebanao.com/wp-content/uploads/2023/09/Office_Cabin_u03.jpg";  
  // or any office space image
```

```
  // Single Office Object
```

```
  const office = {
```

```
    name: "CoWork Hub",
```

```
    rent: 55000,
```

```
    address: "123, Anna Salai, Chennai"
```

```
};
```

```
// List of office objectsa
```

```
const officeList = [
```

```
{
```

```
  name: "CoWork Hub",
```

```
  rent: 55000,
```

```
  address: "123, Anna Salai, Chennai"
```

```
},
```

```
{
```

```
  name: "TechTower",
```

```
  rent: 75000,
```

```
  address: "456, T-Nagar, Chennai"
```

```
},
```

```
{
```

```
  name: "StartupSquare",
```

```
  rent: 62000,
```

```
  address: "789, Velachery, Chennai"
```

```
},
```

```
{
```

```
  name: "BudgetSpace",
```

```
  rent: 40000,
```

```
  address: "101, Guindy, Chennai"
```

```
}
```

```
];
```

```
// Inline CSS logic for rent color
```

```
const getRentStyle = (rent) => {
```

```
  return {
```

```
    color: rent < 60000 ? "red" : "green",
```

```
    fontWeight: "bold"
```

```

    };
  };

  return (
    <div style={{ padding: "20px", fontFamily: "Arial" }}>
      {heading}

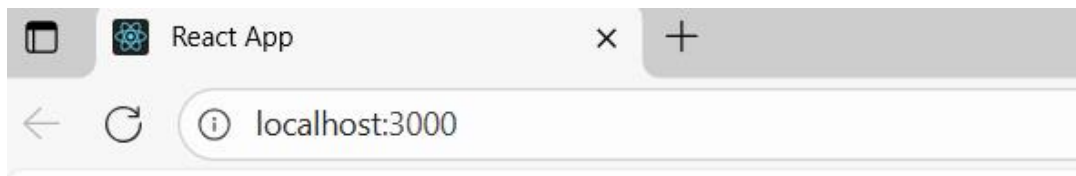
      <img src={imageUrl} alt="Office Space" width="400" style={{ borderRadius: "10px" }} />

      <h2> 🏢 Featured Office:</h2>
      <p>Name: {office.name}</p>
      <p style={getRentStyle(office.rent)}>Rent: ₹{office.rent}</p>
      <p>Address: {office.address}</p>

      <h2> 📋 All Available Offices:</h2>
      <ul>
        {officeList.map((officeItem, index) => (
          <li key={index} style={{ marginBottom: "20px" }}>
            <strong>{officeItem.name}</strong><br />
            <span style={getRentStyle(officeItem.rent)}>Rent: ₹{officeItem.rent}</span><br />
            Address: {officeItem.address}
          </li>
        ))}
      </ul>
    </div>
  );
}

export default App;

```



Office Space Rental App



Featured Office:

Name: CoWork Hub

Rent: ₹55000

Address: 123, Anna Salai, Chennai

All Available Offices:

- **CoWork Hub**
Rent: ₹55000
Address: 123, Anna Salai, Chennai
- **Tech Tower**
Rent: ₹75000
Address: 456, T-Nagar, Chennai
- **StartupSquare**
Rent: ₹62000
Address: 789, Velachery, Chennai
- **BudgetSpace**
Rent: ₹40000
Address: 101, Guindy, Chennai

Hands On 3:

Goal: Demonstrate various event handling techniques, synthetic events, this keyword usage, and form event handling.

App.js

```
import React, { Component } from 'react';
import CurrencyConvertor from './CurrencyConvertor';

class App extends Component {
  constructor(props) {
    super(props);
    this.state = {
      count: 0
    };
  }

  increment = () => {
    this.setState(prevState => ({ count: prevState.count + 1 }));
  };

  sayHello = () => {
    console.log("Hello! This is a static message.");
  };

  increaseHandler = () => {
    this.increment();
    this.sayHello();
  };

  decrement = () => {
    this.setState(prevState => ({ count: prevState.count - 1 }));
  };
}
```

```
};
```

```
sayWelcome = (msg) => {  
  alert(msg);  
};
```

```
handleSyntheticEvent = (e) => {  
  alert("I was clicked");  
  console.log("Synthetic Event:", e);  
};
```

```
render() {  
  return (  
    <div style={{ padding: "20px", fontFamily: "Arial" }}>  
      <h1>React Event Handling Example</h1>  
  
      <h2>Counter: {this.state.count}</h2>  
      <button onClick={this.increaseHandler}>Increase</button>  
      <button onClick={this.decrement}>Decrease</button>
```

```
      <br /><br />
```

```
      <button onClick={() => this.sayWelcome("Welcome to React Events!")}>Say  
Welcome</button>
```

```
      <br /><br />
```

```
      <button onClick={this.handleSyntheticEvent}>Synthetic Event OnPress</button>
```

```
      <br /><br />
```

```
        <CurrencyConvertor />
    </div>

    );
}
}
```

```
export default App;
```

CurrencyConvertor.js

```
import React, { Component } from 'react';
```

```
class CurrencyConvertor extends Component {
  constructor(props) {
    super(props);
    this.state = {
      rupees: "",
      euro: ""
    };
  }
}
```

```
  handleChange = (e) => {
    this.setState({ rupees: e.target.value });
  };
}
```

```
  handleSubmit = () => {
    const rate = 0.011; // 1 INR = 0.011 EUR (sample rate)
    const euroValue = parseFloat(this.state.rupees) * rate;
    this.setState({ euro: euroValue.toFixed(2) });
  };
}
```

```
  render() {
```

```

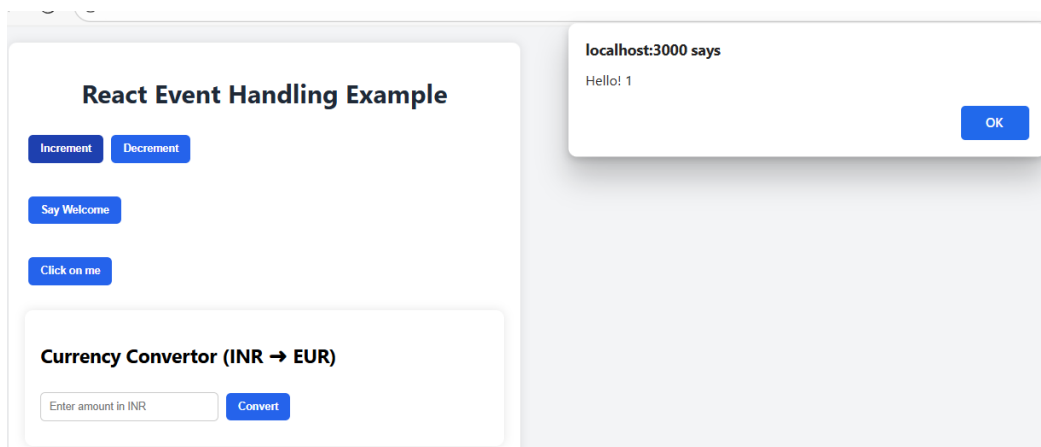
return (
  <div>
    <h2>Currency Convertor (INR → EUR)</h2>
    <input
      type="number"
      placeholder="Enter amount in INR"
      value={this.state.rupees}
      onChange={this.handleChange}
    />
    <button onClick={this.handleSubmit}>Convert</button>

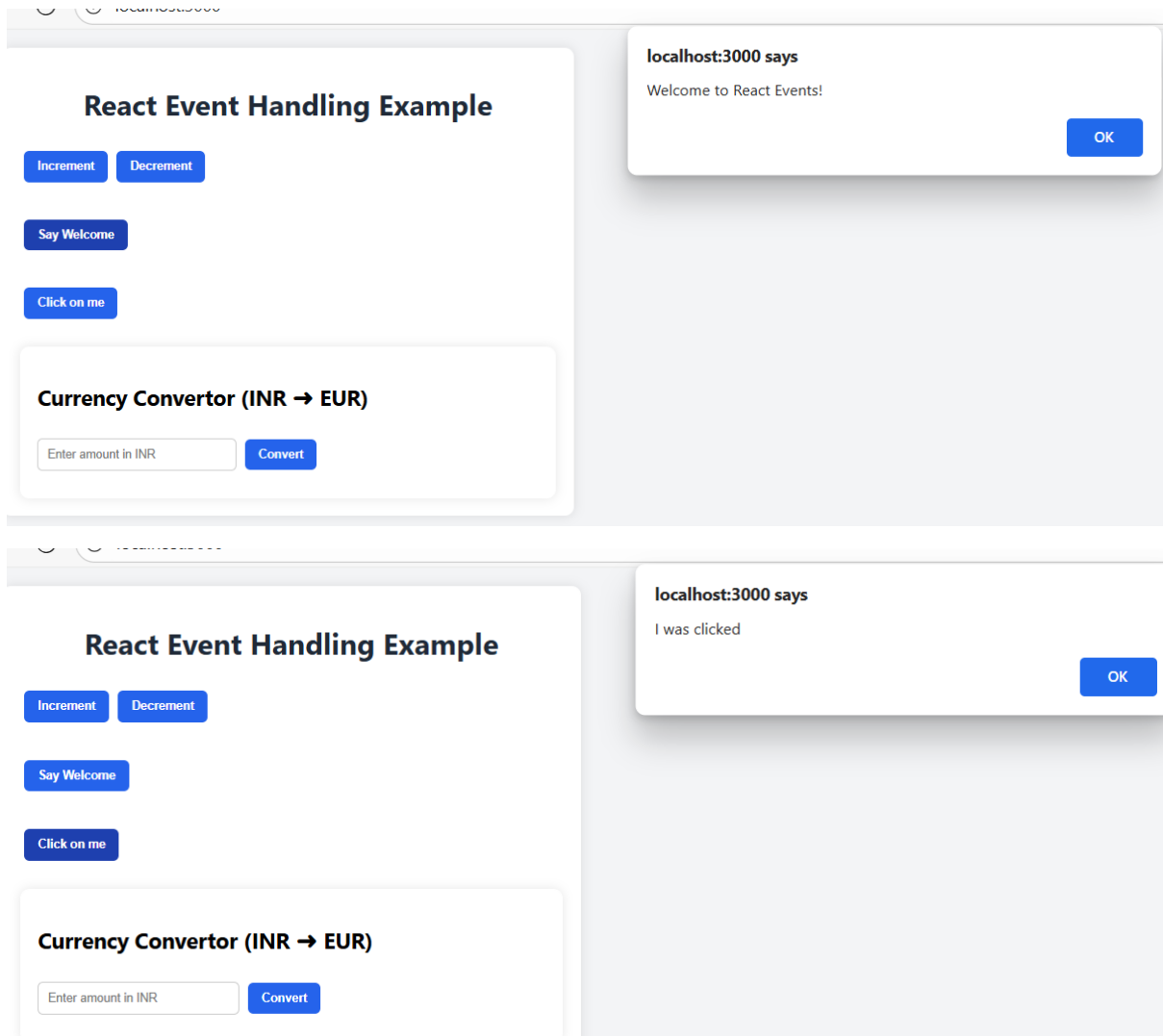
    {this.state.euro && (
      <p>Equivalent in Euro: €{this.state.euro}</p>
    )}
  </div>
);
}
}

```

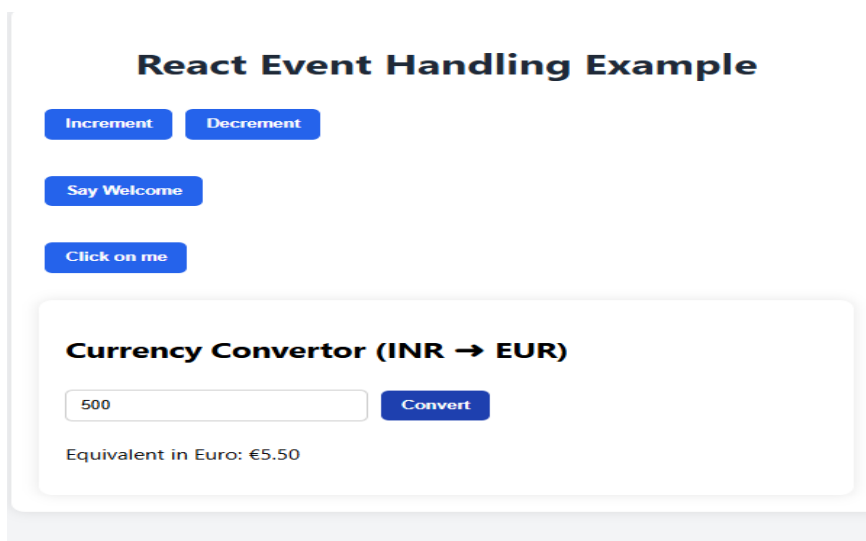
```
export default CurrencyConvertor;
```

Increment” button to increase the value of the counter and “Decrement” button to decrease the value of the counter





Currency Converter



HandsOn 4:

Create a React Application named “ticketbookingapp” where the guest user can browse the page where the flight details are displayed whereas the logged in user only can book tickets.

The Login and Logout buttons should accordingly display different pages. Once the user is logged in the User page should be displayed. When the user clicks on Logout, the Guest page should be displayed.

Guest.js

```
import React from 'react';
import FlightList from './FlightList';

const GuestPage = () => {
  return (
    <div>
      <h2>Welcome Guest! ✈️</h2>
      <p>Please log in to book your tickets.</p>
      <FlightList />
    </div>
  );
};

export default GuestPage;
```

User.js

```
import React from 'react';
import FlightList from './FlightList';
```

```
const UserPage = () => {  
  return (  
    <div>  
      <h2>Welcome, User!  </h2>  
      <p>You can now book tickets.</p>  
      <FlightList />  
      <button>Book Selected Flight</button>  
    </div>  
  );  
};
```

```
export default UserPage;
```

Flight.js

```
import React from 'react';
```

```
const FlightList = () => {  
  const flights = [  
    { id: 1, from: "Chennai", to: "Delhi", time: "10:00 AM" },  
    { id: 2, from: "Mumbai", to: "Bangalore", time: "12:30 PM" },  
    { id: 3, from: "Hyderabad", to: "Kolkata", time: "5:45 PM" }  
  ];  
  
  return (  
    <div>  
      <h3>Available Flights:</h3>  
      <ul>  
        {flights.map(flight => (  

```

```
      <li key={flight.id}>
        {flight.from} → {flight.to} at {flight.time}
      </li>
    )}
  </ul>
</div>
);
};
```

```
export default FlightList;
```

App.js

```
import React, { Component } from 'react';
import GuestPage from './GuestPage';
import UserPage from './UserPage';
import './App.css';
```

```
class App extends Component {
  constructor(props) {
    super(props);
    this.state = {
      isLoggedIn: false
    };
  }

  handleLogin = () => {
    this.setState({ isLoggedIn: true });
  };
}
```

```
handleLogout = () => {  
  this.setState({ isLoggedIn: false });  
};  
  
render() {  
  const { isLoggedIn } = this.state;  
  
  let page;  
  if (isLoggedIn) {  
    page = <UserPage />;  
  } else {  
    page = <GuestPage />;  
  }  
  
  return (  
    <div className="container">  
      <h1> ✈ Ticket Booking App</h1>  
  
      {isLoggedIn ? (  
        <button onClick={this.handleLogout}>Logout</button>  
      ) : (  
        <button onClick={this.handleLogin}>Login</button>  
      )}  
  
      <hr />  
  
      {page}
```

```
        </div>
    );
}
}
```

```
export default App;
```

App.css

```
.container {
  padding: 20px;
  font-family: Arial;
  max-width: 600px;
  margin: auto;
}
```

```
button {
  padding: 10px 20px;
  margin: 10px 0;
  font-weight: bold;
  background-color: #1e90ff;
  color: white;
  border: none;
  border-radius: 6px;
  cursor: pointer;
}
```

```
button:hover {
  background-color: #0c6ed7;
```

}

Ticket Booking App

Login

Welcome Guest! 

Please log in to book your tickets.

Available Flights:

- Chennai → Delhi at 10:00 AM
- Mumbai → Bangalore at 12:30 PM
- Hyderabad → Kolkata at 5:45 PM

Ticket Booking App

Logout

Welcome, User! 

You can now book tickets.

Available Flights:

- Chennai → Delhi at 10:00 AM
- Mumbai → Bangalore at 12:30 PM
- Hyderabad → Kolkata at 5:45 PM

Book Selected Flight

Hands On 5:

Create a React App named “bloggerapp” in with 3 components.

1. Book Details
2. Blog Details
3. Course Details

Implement this with as many ways possible of Conditional Rendering.

Books.js

```
import React from 'react';
```

```
const BookDetails = () => {
```

```
  const books = [
```

```
    { id: 1, title: "Atomic Habits", author: "James Clear" },
```

```
    { id: 2, title: "Deep Work", author: "Cal Newport" },
```

```
    { id: 3, title: "Ikigai", author: "Héctor García" }
```

```
  ];
```

```
  return (
```

```
    <div>
```

```
      <h2>  Book Details</h2>
```

```
      <ul>
```

```
        {books.map(book => (
```

```
          <li key={book.id}>
```

```
            <strong>{book.title}</strong> - {book.author}
```

```
          </li>
```

```
        ))}
```

```
      </ul>
```

```
    </div>
```

```
);  
};
```

```
export default BookDetails;
```

```
Course.js
```

```
import React from 'react';
```

```
const CourseDetails = () => {
```

```
  const courses = [
```

```
    { id: 1, name: "React Basics", instructor: "Jane" },
```

```
    { id: 2, name: "Full Stack Development", instructor: "Mark" },
```

```
    { id: 3, name: "Data Structures", instructor: "Lisa" }]
```

```
};
```

```
  return (
```

```
    <div>
```

```
      <h2>🎓 Course Details</h2>
```

```
      <ul>
```

```
        {courses.map(course => (
```

```
          <li key={course.id}>
```

```
            {course.name} - {course.instructor}
```

```
          </li>
```

```
        )]}
```

```
      </ul>
```

```
    </div>
```

```
  );
```

```
};
```

```
export default CourseDetails;
```

Blog.js

```
import React from 'react';
```

```
const BlogDetails = () => {
```

```
  const blogs = [
```

```
    { id: 1, title: "Why React is Awesome", author: "Alex" },
```

```
    { id: 2, title: "Learning JavaScript", author: "Maria" },
```

```
    { id: 3, title: "CSS Tips & Tricks", author: "John" } 
```

```
  ];
```

```
  return (
```

```
    <div>
```

```
      <h2>  Blog Details</h2>
```

```
      <ul>
```

```
        {blogs.map(blog => (
```

```
          <li key={blog.id}>
```

```
            <strong>{blog.title}</strong> - by {blog.author}
```

```
          </li>
```

```
        )))
```

```
      </ul>
```

```
    </div>
```

```
  );
```

```
};
```

```
export default BlogDetails;
```

App.js

```
import React, { Component } from 'react';
```

```
import BookDetails from './BookDetails';
import BlogDetails from './BlogDetails';
import CourseDetails from './CourseDetails';
import './App.css';

class App extends Component {
  constructor(props) {
    super(props);
    this.state = {
      selected: 'books' // default view
    };
  }

  renderComponent() {
    // if-else method
    if (this.state.selected === 'books') {
      return <BookDetails />;
    } else if (this.state.selected === 'blogs') {
      return <BlogDetails />;
    } else if (this.state.selected === 'courses') {
      return <CourseDetails />;
    } else {
      return <p>Please select a section above.</p>;
    }
  }

  render() {
    const { selected } = this.state;

    // element variable
```

```
let heading;

switch (selected) {
  case 'books':
    heading = "Books";
    break;
  case 'blogs':
    heading = "Blogs";
    break;
  case 'courses':
    heading = "Courses";
    break;
  default:
    heading = "";
}

return (
  <div className="container">
    <h1> Blogger App</h1>

    <div className="tabs">
      <button onClick={() => this.setState({ selected: 'books' })}>Book Details</button>
      <button onClick={() => this.setState({ selected: 'blogs' })}>Blog Details</button>
      <button onClick={() => this.setState({ selected: 'courses' })}>Course Details</button>
    </div>

    <h3>Currently viewing: {heading}</h3>

    {/* Conditional render using function (if-else) */}
    {this.renderComponent()}
```

```

    { /* Additional ternary operator example */ }

    <p style={{ marginTop: "10px" }}>
      {selected === 'books' ? "Explore popular books." :
        selected === 'blogs' ? "Check out trending blogs." :
        selected === 'courses' ? "Available courses for you." : ""}
    </p>

    { /* Logical && operator */ }
    {selected && <p>[{selected.toUpperCase()}] section selected.</p>}
  </div>

);
}
}

```

```
export default App;
```

App.css

```

.container {
  font-family: Arial;
  padding: 20px;
  max-width: 700px;
  margin: auto;
}

.tabs button {
  margin-right: 10px;
  padding: 10px;
  background-color: #1e88e5;
  color: white;
}

```

```
border: none;

border-radius: 5px;

cursor: pointer;
}

.tabs button:hover {
  background-color: #1565c0;
}
```



Blogger App

Book Details

Blog Details

Course Details

Currently viewing: Books



Book Details

- **Atomic Habits** - James Clear
- **Deep Work** - Cal Newport
- **Ikigai** - Héctor García

Explore popular books.

[BOOKS] section selected.



Blogger App

[Book Details](#)[Blog Details](#)[Course Details](#)

Currently viewing: **Blogs**



Blog Details

- **Why React is Awesome** - by Alex
- **Learning JavaScript** - by Maria
- **CSS Tips & Tricks** - by John

Check out trending blogs.

[BLOGS] section selected.



Blogger App

[Book Details](#)[Blog Details](#)[Course Details](#)

Currently viewing: **Courses**



Course Details

- React Basics - Jane
- Full Stack Development - Mark
- Data Structures - Lisa

Available courses for you.

[COURSES] section selected.